

Vinh Ly

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Summary

Staff-level engineer with **18 years of experience** building and operating production web systems at YouTube, Google, and Microsoft. Solid full-stack background spanning **Python**, **Node.js**, and **React**, with strong opinions on API design, developer tooling, and writing code that doesn't need emergency paging at 2 am. Currently interested in roles where I can own the full web stack, from backend services to the front end that users actually see.

Core Skills

Languages: Python, Java, C#, TypeScript, Go

Frameworks: React, Node.js, Flask, Spring Boot

Databases: PostgreSQL, MySQL, Redis, MongoDB

Infra & Tooling: Docker, Kubernetes, GitHub Actions, Terraform, GCP, AWS

Methodology: REST API design, CI/CD pipelines, unit and integration testing, code review culture

Work Experience

YouTube | Staff Software Engineer | Remote | 02/2025 - Present

- Own infrastructure reliability for backend services supporting the video platform. **Python** microservices on Kubernetes, handling traffic at YouTube scale. P99 SLA stayed above 99.95% through two high-profile live events this year.
- Migrated a legacy batch-processing pipeline to a streaming architecture using Kafka and **Python** async workers. Reduced end-to-end processing latency from around 45 minutes to under 3 minutes for the data team.

Whatnot | Staff Software Engineer | Remote | 05/2024 - 02/2025

- Built and shipped backend services for a live commerce marketplace handling real-time bidding and payments. Worked across the full stack in **Python** and **Node.js**, integrating with Stripe and managing Postgres at scale.
- Rewrote a slow inventory sync service. Cut average sync time from 20 minutes to under 90 seconds for high-volume sellers, which directly reduced support tickets from the top 10% of merchants.

Google | Staff Software Engineer + Manager | Hybrid | 03/2015 - 04/2024

- Managed a team of 8 engineers covering infrastructure and product areas across YouTube and Google Cloud. Ran performance reviews, scoped quarterly roadmaps, and handled incident rotation for a system serving thousands of internal Google teams.
- Led the migration of internal teams from gRPC to REST with OpenAPI specifications. The migration ran over 18 months with 0 user-facing downtime, and onboarding time for new API consumers dropped from about 2 weeks to 2 days.
- Started a code review dashboard as a side project on the team. It tracked review turnaround, churn rate, and test coverage across repos. Adopted by 60+ teams within a year, and I folded it into the team's sprint planning routine.

Google | Software Engineer, Senior Software Engineer | Remote | 02/2011 - 02/2015

- Worked on the internal developer platform that managed CI/CD pipelines and deployment tooling for hundreds of Google teams. Built backend services in **Java** and a CLI tool that automated environment setup. Setup time for new engineers dropped from several weeks to under a day; the CLI got thousands of monthly active users inside Google.

- Authored an internal testing framework that enforced minimum unit and integration test coverage per service. Teams that adopted it saw production incidents drop by about 40% over the following year.

***Microsoft* | Software Design Engineer II | Hybrid | 10/2007 - 12/2011**

- Worked on Visual Studio extensions and the Visual Studio Online REST API, primarily in C# and ASP.NET. Contributed to database schema design and query performance tuning for the TFS data warehouse. Added covering indexes that brought some reporting queries down from over 2 seconds to roughly 80 milliseconds.
- Set REST API design standards and best practices for the Visual Studio Online web services. The patterns established then carried forward into what eventually became Azure DevOps.

***Microsoft* | Software Design Engineer Intern | Hybrid | 06/2006 - 08/2006**

- Built a prototype internal reporting tool for the Visual Studio product team during a summer internship. The tool got handed to a full-time engineer after the internship ended and was in active use for over a year before the team replaced it with a production system.

Education

BS Computer Science | University of Texas at Austin | 2007